

External Validation Report

Fixed brain MRI research segmentation prototype | 60-case external cohort

RESEARCH ONLY

This report evaluates segmentation metadata and expert-mask agreement. It is not clinical validation, a diagnosis, or treatment advice.

Release decision

Freeze the current checkpoint for the research prototype. Do not retrain or tune on this external cohort. The aggregate result is strong enough for a working demonstration, while the observed miss requires explicit expert-review safeguards.

Mean Dice	Median Dice	Median HD95	Median case time
0.878	0.923	4.47 mm	4.20 s

All external metrics

Metric	Mean	95% CI for mean	Median	P05	P95
Dice	0.878	0.838 to 0.909	0.923	0.702	0.965
IoU	0.802	0.759 to 0.840	0.858	0.541	0.933
Precision	0.887	0.841 to 0.924	0.952	0.657	0.987
Recall	0.879	0.840 to 0.909	0.927	0.691	0.968
HD95 (mm)	12.59	8.56 to 17.24	4.47	1.94	51.34

Outcome distribution and weak-case review

Dice distribution: 35/60 at or above 0.90; 19/60 from 0.75 to 0.90; 5/60 from 0.50 to 0.75; and 1/60 below 0.50. The below-0.75 review is descriptive and post hoc, not a promotion gate.

Six weak cases were reviewed using anonymized metrics only: 1 empty prediction, 2 substantial oversegmentations, and 3 other overlap or boundary errors. The empty prediction was reproducible and occurred against a small 1,026-voxel reference mask, showing a model miss rather than a pipeline crash.

Retraining decision

Failures are mixed rather than one consistent correctable mode; the external cohort remains evaluation-only. Use development data only to test stronger small-lesion sampling and preserve the current model as rollback. Evaluate any new candidate once on a new untouched cohort; do not reuse these 60 cases for selection.

Frozen provenance

Release tag	prototype-v0.1.0
Model	glioma-segresnet-20260828
Checkpoint SHA-256	121422a861bbe7affaa5e161058e69eea737b2390651c3c03ea20256969e99e5
Plan SHA-256	5f8e5a9009ba81e71156dd2cc535d1446fe0d6aba8c09bb996fe59b01ce43d63
Evaluator SHA-256	42cb6e4c62fd430c56b329cde7bec98482207ad23759bb933396047d4fa99eaf
Aggregate JSON SHA-256	c50f15074f4e337e1826f7fa9230b2e3ca642d62ebae9c15545843a58e931ac6
Dataset revision	76608935145af0fb74a74b68f041471aa494f0f6

Method and limitations

The fixed threshold and checkpoint were run once over all 60 accepted BraTS-SSA cases. Whole-lesion Dice, IoU, precision, recall, and physical-spacing HD95 were computed against expert masks. Case-level bootstrap confidence intervals used 10,000 fixed-seed replicates. Native case identifiers and paths are excluded from this report.

This cohort does not prove clinical safety, calibration, generalization to every scanner or population, or absence of disease when the output is empty. Expert review remains required. Any future candidate requires development-only iteration and a new untouched evaluation cohort.

Benchmark generated 2026-09-04T18:00:48.875355+00:00